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Java SE 27 & JDK 27

Class ClassValue<T>

java.lang.Object
    java.lang.ClassValue<T>

Type Parameters:

T - the type of the associated value

```
public abstract class ClassValue<T>
extends Object
```

Lazily associate a computed value with any `Class` object. For example, if a dynamic language needs to construct a message dispatch table for each class encountered at a message send call site, it can use a `ClassValue` to cache information needed to perform the message send quickly, for each class encountered.

The basic operation of a `ClassValue` is `get`, which returns the associated value, initially created by an invocation to `computeValue`; multiple invocations may happen under race, but exactly one value is associated to a `Class` and returned.

Another operation is `remove`: it clears the associated value (if it exists), and ensures the next associated value is computed with input states up-to-date with the removal.

For a particular association, there is a total order for accesses to the associated value. Accesses are atomic; they include:

- A read-only access by `get`
- An attempt to associate the return value of a `computeValue` by `get`
- Clearing of an association by `remove`

A `get` call always include at least one access; a `remove` call always has exactly one access; a `computeValue` call always happens between two accesses. This establishes the order of `computeValue` calls with respect to `remove` calls and determines whether the results of a `computeValue` can be successfully associated by a `get`.

Since:

1.7

Constructor Summary

Constructors

Modifier	Constructor	Description

protected	**ClassValue**() Sole constructor.

Method Summary

All Methods	Instance Methods	

Modifier and Type	Method	Description
protected abstract T	**computeValue**(Class<?> type)	Computes the value to associate to the given Class.
T	**get**(Class<?> type)	Returns the value associated to the given Class.
void	**remove**(Class<?> type)	Removes the associated value for the given Class and invalidates all out-of-date computations.

Methods declared in class Object

clone, equals, finalize, getClass, hashCode, notify, notifyAll, toString, wait, wait, wait

Constructor Details

ClassValue

protected ClassValue()

Sole constructor. (For invocation by subclass constructors, typically implicit.)

Method Details

computeValue

protected abstract T computeValue(Class<?> type)

Computes the value to associate to the given Class.

This method is invoked when the initial read-only access by get finds no associated value.

If this method throws an exception, the initiating `get` call will not attempt to associate a value, and may terminate by returning the associated value if it exists, or by propagating that exception otherwise.

Otherwise, the value is computed and returned. An attempt to associate the return value happens, with one of the following outcomes:

- The associated value is present; it is returned and no association is done.
- The most recent `remove` call, if it exists, does not happen-before (JLS §17.4.5) the finish of the `computeValue` that computed the value to associate. A new invocation to `computeValue`, which that `remove` call happens-before, will re-establish this happens-before relationship.
- Otherwise, this value is successfully associated and returned.

API Note:

A `computeValue` call may, due to class loading or other circumstances, recursively call `get` or `remove` for the same `type`. The recursive `get`, if the recursion stops, successfully finishes and this initiating `get` observes the associated value from recursion. The recursive `remove` is no-op, since being on the same thread, the `remove` already happens-before the finish of this `computeValue`; the result from this `computeValue` still may be associated.

Parameters:

`type` - the `Class` to associate a value to

Returns:

the newly computed value to associate

See Also:

`get(Class)`, `remove(Class)`

get

```
public T get(Class<?> type)
```

Returns the value associated to the given `Class`.

This method first performs a read-only access, and returns the associated value if it exists. Otherwise, this method tries to associate a value from a `computeValue` invocation until the associated value exists, which could be associated by a competing thread.

This method may throw an exception from a `computeValue` invocation. In this case, no association happens.

Parameters:

`type` - the `Class` to retrieve the associated value for

Returns:

the value associated to the given `Class`

Throws:

`NullPointerException` - if the argument is `null`

See Also:

`remove(Class)`, `computeValue(Class)`

remove

```
public void remove(Class<?> type)
```

Removes the associated value for the given `Class` and invalidates all out-of-date computations. If this association is subsequently accessed, this removal happens-before (JLS §17.4.5) the finish of the `computeValue` call that returned the associated value.

Parameters:

`type` - the type whose class value must be removed

Throws:

`NullPointerException` - if the argument is `null`

Report a bug or suggest an enhancement
For further API reference and developer documentation see the Java SE Documentation, which contains more detailed, developer-targeted descriptions with conceptual overviews, definitions of terms, workarounds, and working code examples. Other versions.
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